

K. J. Somaiya College of Engineering, Mumbai-77

End Semester Exam

December 2020

Max. Marks: 30

Duration: 1 Hr. 15 Min. (For attempting questions) + 15 min. (uploading)

Class: TY

Name of the Course: SOFT COMPUTING

Course Code: 2UCE504

Semester: V

Branch : COMP

Instructions:

- (1) **All questions are compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

Question No.		Max. Marks																																								
Q.1-a)	Explain defuzzification methods OR Explain different types of activation functions	5																																								
Q.1-b)	What is learning in neural network? Distinguish between supervised and unsupervised learning.	5																																								
Q.2	Explain error back propagation algorithm with the help of diagram. OR Explain genetic algorithm with the help of an example.	10																																								
Q3-a)	High speed rail monitoring devices sometimes make use of sensitive sensors to measure the deflection of the earth when a rail car passes. These deflections are measured with respect to some distance from the rail car and, hence are actually very small angles measured in micro-radians. Let a universe of deflection be $A = [1, 2, 3, 4]$ where A is the angle in micro-radians, and let a universe of distance be $D = [1, 2, 5, 7]$ where D is distance in feet, suppose a relation between these two parameters has been determined as follows <table><tr><td></td><td>D_1</td><td>D_2</td><td>D_3</td><td>D_4</td></tr><tr><td>A_1</td><td>1</td><td>0.3</td><td>0.1</td><td>0</td></tr><tr><td>A_2</td><td>0.2</td><td>1</td><td>0.3</td><td>0.1</td></tr><tr><td>A_3</td><td>0</td><td>0.7</td><td>1</td><td>0.2</td></tr><tr><td>A_4</td><td>0</td><td>0.1</td><td>0.4</td><td>1</td></tr></table> Now let a universe of rail car weights be $W = [1, 2]$, where W is the weight in units of 100,000 pounds. Suppose the fuzzy relation of W to A is given by, <table><tr><td></td><td>W_1</td><td>W_2</td></tr><tr><td>A_1</td><td>1</td><td>0.4</td></tr><tr><td>A_2</td><td>0.5</td><td>1</td></tr><tr><td>A_3</td><td>0.3</td><td>0.1</td></tr><tr><td>A_4</td><td>0</td><td>0</td></tr></table>		D_1	D_2	D_3	D_4	A_1	1	0.3	0.1	0	A_2	0.2	1	0.3	0.1	A_3	0	0.7	1	0.2	A_4	0	0.1	0.4	1		W_1	W_2	A_1	1	0.4	A_2	0.5	1	A_3	0.3	0.1	A_4	0	0	5
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	Using these two relations, find the relation $R^T \circ S$. (a) Using max-min composition. (b) Using max-product composition.	
Q.3-b)	Develop graphical representation of membership function to describe linguistic variables "cold", "warm" and "hot". The temperature ranges from 0°C to 100°C. Also show plot for "cold and warm" and "warm or hot" temperature.	5